Annie: User Manual

Annie is an accounts-payable (AP) tool for **three-way matching**. It reads purchase orders or purchase requisitions (PO/PRs) and supplier invoices, checks them, matches each invoice to its PO/PR lines, and tracks goods receipts and service entries (GR/SES) until the case is complete and can be archived.

- **Live version (no install needed):** https://anniev5-production.up.railway.app
- **Source code:** https://github.com/ryanteoyx/anniev5 (branch `ui-polish`)

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1. What is in the package

Path	What it is
`app.py`	The web interface (Streamlit)
`annie_core.py`	The engine: reading documents, checks, matching, storage, email intake
`tour.py`	The ▶ **Play demo** guided tour
`demo.py`, `demo_screenshots/`	The **Slides** slideshow
`samples/`	28 sample PO/PRs and invoices; loaded automatically on first start
`requirements.txt`	The Python packages Annie needs, with tested versions
`.env.example`	Optional settings (admin PIN, email inbox); copy to `.env` to use
`docs/BUSINESS_LOGIC.md`	Every rule, score and threshold, with worked examples
`docs/ARCHITECTURE.md`	Components, data model and deployment
`tests/`	Automated tests
`Dockerfile`	Container build (used for the live version)
`deploy/`	Optional setup for a self-hosted server

Annie creates `annie_v4.db` (the database) and `storage/` (uploaded files) when it runs. Both can be deleted to start fresh.

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2. Requirements

Python	**3.11** (3.12 also works). Avoid very new versions such as 3.14: some packages have no ready-made install files for them yet.
Disk space	About 1.5 GB for the Python packages (the OCR engine is the largest part)
Internet	Needed once, to install the packages
Operating system	macOS, Windows or Linux
Browser	Any modern browser (Chrome, Edge, Safari, Firefox)
Optional	Tesseract OCR as a second OCR engine (`brew install tesseract` on macOS). Not required: RapidOCR is installed with the packages.

The Python packages (from `requirements.txt`):

Package	Used for
streamlit	Web interface
pandas, numpy, openpyxl	Tables, CSV and Excel files
pdfplumber, PyMuPDF	Reading PDFs; rendering pages for OCR and the signature check
python-docx	Word files
Pillow, pillow-heif	Images, including iPhone HEIC photos
rapidocr, onnxruntime	OCR for scanned documents
pytesseract	Optional second OCR engine (needs Tesseract installed)
python-dotenv	Reading settings from `.env`

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3. Install

Unzip the package, open a terminal in the unzipped `anniev5` folder, then:

macOS / Linux

```
python3.11 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Windows (Command Prompt or PowerShell)

```
py -3.11 -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
```

The install takes a few minutes. Check it worked:

```
python tests/test_findings.py
```

It should end with `11 passed, 0 failed`.

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4. Start and stop Annie

With the environment active (step 3):

```
streamlit run app.py
```

Annie opens in your browser at **http://localhost:8501**. On the first start it loads the 28 sample documents, so the screens are not empty.

To stop Annie, press **Ctrl + C** in the terminal. Next time, activate the environment again (`source .venv/bin/activate`, or `.venv\Scripts\activate` on Windows) and run `streamlit run app.py`.

Docker instead of Python (if Docker is installed):

```
docker build -t annie .
docker run -p 8501:8501 annie
```

5. How a document moves through Annie

```
Upload or email
      |
      v
Module 1: Intake ---- missing fields, unreadable, duplicate ----> Intake Review (a person c
      |
      v
Module 2: Match
  Stage 1  Open cases       invoice waiting for its PO/PR, or held by a check
  Call-Offs                 PO/PR with no amount: matched by hand
  Stage 2  Pending GR/SES   matched; waiting for what was received
  Stage 3  Complete         receipts cover the invoice; ready to post
      |
      v
Complete Archive (ZIP package per case)
```

6. Using each page

The pages are in the left sidebar, in the order of the AP flow.

Dashboard

Counts for every stage. Click a tile to jump to that list.

Submit Document

- **Upload documents:** choose PDF, image, DOCX, TXT, CSV or XLSX files (up to 20 MB each) and click **Process documents**.
- **Email inbox:** if an inbox is set up (section 8), **Check inbox now** fetches new attachments. Annie also checks it automatically.
- **Results this session** shows what happened to each file: registered, matched, waiting, sent to Intake Review, or rejected as a duplicate.

Intake Review

Documents Module 1 could not pass on: missing mandatory fields (for example, no invoice number or date), unreadable, or not a PO/PR or invoice. The original file is shown next to the list of what is missing. - **Re-extract from file:** read the file again (after a fix to the file or the settings). - **Dismiss:** remove the document completely. It can be submitted again later.

Processing

Four tabs.

Stage 1 - Open cases. Invoices that could not be matched automatically, each with the reason (for example, "PO/PR not yet received", "unit price 7.50 is 200% above the PO/PR price"). Below them, every PO/PR that still has a balance, with ordered, billed and received amounts (**Download CSV** exports this list). Click **Open** on an invoice to: - see the invoice and its PO/PR side by side; - **Retry automatic allocation**; - **Match manually:** choose a PO/PR line for each invoice line (one line can be split over several PO/PR lines), type a reason, and confirm. The reason goes into the audit log; - **Dismiss invoice (delete):** tick the confirmation box. Any billing on the PO/PR is reversed and the file can be uploaded again later.

A waiting invoice is retried automatically whenever a new PO/PR arrives.

Call-Offs. PO/PRs with no amount (for example, "pest control as required"). Pick the invoices that belong to the call-off, type a reason and click **Confirm**.

Stage 2 - Pending GR/SES. Matched invoices waiting for proof of delivery or service. Open a case to see how each invoice line was allocated, then fill in the GR/SES form (what was actually received in this batch) and click **Submit GR/SES**. Partial deliveries are fine; the case completes when receipts cover the invoice. **Unmatch (send back to Stage 1)** undoes the match, with a reason.

Stage 3 - Complete. Fully received cases. Open one and click **Post** to archive it. Post downloads a ZIP with the documents and a summary. It does **not** post anything to an accounting system.

Complete Archive

Posted cases. **Open** to view one, **Package** to download its ZIP again.

Documents

One row per document: how it arrived, what Module 1 decided, and what Module 2 did with it. Filter and sort with the controls above the table.

Audit Log

Every event: received, rejected, matched, manually matched (with the reason), GR/SES submitted, posted.

Controls

- **Demo data → Reset & load demo data:** back to the starting sample set. Needs the admin PIN.
- **Inbox monitor:** status of automatic email fetching.
- **Module 1 status:** OCR engine, reference samples, email settings, and the matching thresholds.
- **Danger zone → Reset database (empty):** removes everything, for example before uploading live during a demo. Needs the admin PIN.

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7. Demo mode

- **▶ Play demo** (top right): a guided tour of 23 stops. A cursor clicks through the pages and zooms in on each feature. **Next** or **Space** moves on, ← goes back, **F** is full screen, **Esc** leaves. It uses its own copy of the demo data, so it changes nothing. Keep the browser zoom at 100%.
- **Slides:** the same features as a slideshow of screenshots.

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8. Settings (optional)

Copy `.env.example` to `.env` in the same folder and fill in what you need, then restart Annie.

Setting	What it does
`ADMIN_PIN`	Unlocks the two reset buttons on Controls. Without it they stay locked.
`DEMO_AUTOLOAD`	`1` = load the samples into an empty database (default); `reset` = reload them on every start; `0` = never

`EMAIL_HOST`, `EMAIL_ADDRESS`, `EMAIL_PASSWORD`	Turn on email intake (IMAP). The mailbox is read-only: mail is never marked as read or deleted. Use an app password, not your normal password.
`EMAIL_PORT`, `EMAIL_MAILBOX`, `EMAIL_SUBJECT_FILTER`, `EMAIL_LOOKBACK_DAYS`, `POLL_SECONDS`, `EMAIL_AUTO_FETCH`	Fine-tuning for email intake (defaults: 993, INBOX, none, 30 days, 60 seconds, on)

Never put `.env` in a shared folder or on GitHub: it can contain passwords.

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9. Matching rules in brief

An invoice line is matched **automatically** only when all of these hold (details in `docs/BUSINESS_LOGIC.md`):

Check	Rule
Vendor	Names at least **90%** alike, ignoring endings such as Pte Ltd, Sdn Bhd, LLC
Currency	Same currency (an unstated currency does not block)
Line score	At least **0.80** (60% description, 40% unit price; +0.25 when the invoice cites the PO/PR)
Price (goods)	Unit price at most **5%** above the PO/PR price
Line total (goods)	At most quantity × unit price + 0.5% (+0.05 for rounding)
Item (goods)	Description at least **0.40** similar to the PO/PR item
Balance	Enough quantity or amount left on the PO/PR line
Whole invoice	Every line must fit; otherwise the invoice waits in Stage 1 with the reason

Anything that fails waits for a person, who can match it manually with a typed reason.

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10. Run the tests

```
python tests/test_findings.py      # matching and intake rules (11 tests)
python tests/test_tour.py          # the Play demo data (2 tests)
python tests/sample_regression.py  # the 28 sample documents give the expected 17 cases
```

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11. Troubleshooting

Problem	Fix
`pip install` fails on `onnxruntime` or `PyMuPDF`	Use Python 3.11 or 3.12 (step 2). Check with `python --version`.
`No space left on device` during install	Free about 1.5 GB of disk space and run `pip install -r requirements.txt` again.
`streamlit: command not found`	The environment is not active: run `source .venv/bin/activate` (Windows: `.venv\Scripts\activate`).
Port 8501 already in use	`streamlit run app.py --server.port 8502`, then open http://localhost:8502
Scanned documents come out empty	The first scan downloads the OCR models (needs internet once). Clear, upright scans work best.

Want to start again	Stop Annie, delete `annie_v4.db` and the `storage/` folder, start again.
Reset buttons are greyed out	Set `ADMIN_PIN` in `.env` (section 8) and restart.
The Play demo looks cut off	Set the browser zoom to 100% (Cmd/Ctrl + 0).

12. Limitations

- Tested on synthetic sample documents only, not on real company data.
- **Post** archives a case as a ZIP; nothing is sent to Xero or any other accounting system.
- One company, one currency per match, no user logins or roles.
- The signature check on PO/PR forms looks for ink in the signature area; it is not proof of approval.
- Duplicate checks catch the same file, the same content and repeated reference numbers; a duplicate charge with a new invoice number and different wording needs a person.
- Thresholds are prototype settings and should be tuned on real data.